



The Wrong Number

Quick facilitation notes & answer key

Goal

Students fix a wrong-VALUE bug in one of two parallel scripts (C3.2 alter + C3.1 write). The goal is 'both reach 4'; Bit is correct, but Pixel's 'forward 3' falls one short. Students locate the buggy script, work out the correct number (forward 4), and write the fix. Altering a value and re-checking the combined goal is the core debugging skill here.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a star on square 4; run Pixel from 0 and count the squares to the star to find the right number.

Run it (about 20 min)

1. I DO: state the goal — 'both reach 4.' Run Bit: 1 forward 3 → 4 (correct).
2. WE DO: run Pixel: 0 forward 3 → 3 (short by 1). 'Pixel is the bug — its number is too small.'
3. YOU DO: students count 0 to 4 (four squares), so the move should be forward 4; they write the fix and confirm Pixel reaches 4.
4. Connect: 'Right block, wrong number — count to the goal to fix the value.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 1 → forward 3 → 4 (reaches goal). Pixel: 0 → forward 3 → 3 (misses; Pixel is the buggy one).

Ex 2 — Pixel starts at 0 and needs 4, so the move should be forward 4.

Ex 3 — fixed move: forward 4 → Pixel lands on 4. Now BOTH reach the goal 4.

Fix note: only the number changes; the block type (forward) and Bit's script stay the same.

Watch for & differentiate

Common error: changing Bit's script — Bit already reaches 4; the wrong number is in Pixel's move.

Common error: over-correcting to forward 5 (overshoots to 5) — count carefully from the home square.

Simplify: have students put a finger on Pixel's home (0) and hop to the star, counting the hops (4).

Extend: ask what Pixel's move would need to be if its home were 1 instead of 0 (forward 3).

Success indicator: the child fixes Pixel to forward 4 and confirms both sprites reach 4.